

Week - 13 - DSA I

- Data Structure ✓
- Algorithms ✓
- Memory allocation ✓
- memory leak ✓
- Complexity analysis ✓
- Asymptotic analysis ✓
- Big-O notation ✓
- Concepts of array ✓
- Concepts of linked list ✓
- Singly linked list ✓
- Doubly linked list ✓
- Concepts of string ✓
- Binary search ✓
- Linear search ✓
- Recursion ✓ (string reverse)
- application of all structures ✓
- Linear vs non-Linear ✓
- + Types of ds ✓
- Types of memory allocation ✓
- ✓ Contiguous and non contiguous
- ✓ Garbage collection and working
- ✓ Adv and Disadv of linked list
- ✓ mutable and immutable
- Hierarchical DS
- Tagged array.
- Adv and Disadv of recursion
- List functions.
- Virtual memory.
- ✓ circular reference.
- ✓ heterogeneous array.
- ✓ (set, insert, delete, init, sort, ascending, descending, linked list, Reverse doubly.)
- null termination.
- ✓ LF, CR LF
- ✓ Unical.

Week 14 . Data structure II

- ✓ Bubble sort $O(n^2)$
 - ✓ Insertion sort $O(n^2)$
 - ✓ Selection sort $O(n^2)$
 - ✓ Quick sort $O(n \log n)$, $O(n^2)$
 - ✓ Merge sort $O(n \log n)$
 - ✓ Stack (types) adv/dis
 - ✓ Queue (types) adv/dis
 - ✓ Push and pop in stack
 - ✓ Enqueue, Dequeue in Queue
 - ✓ Hash tables
 - ✓ Applications all structures
 - ✓ Peek
 - ✓ Stable and unstable
 - ✓ deterministic and undeterministic
 - ✓ Stack underflow and stack overflow
 - ✓ divide and conquer strategy.
 - ✓ adv & dis using stack and queue.
 - ✓ double ended queue.
 - ✓ Hash Functions.
 - ✓ Hash values.
 - ✓ collision.
 - ✓ method to prevent collision.
 - ✓ chaining
 - ✓ linear probing
 - ✓ Quadratic, &
 - ✓ double hashing
 - ✓ Load factor.
 - ✓ array vs hashtable
 - ✓ adv and disadv of hashtable
 - ✓ ~~Stack~~ Bubble sort conditions
 - ✓ Lifo queue
 - ✓ Priority queue
- ### Practicals

 - ✓ reverse string in stack
 - ✓ find long char in string
 - ✓ String Reverse using stack
 - ✓ "aabcdbbefgccc" output: a:1, b:4, c:3
 - ✓ stack second largest
 - ✓ frequency of string using hash map.
 - ✓ delete middle element of stack
 - ✓ valid parenthesis {([)]}
 - ✓ queue reverse in recursion.
 - ✓ find duplicates using hash table

Week - 15 - Data structures III

- ✓ Concepts of tree.
- ✓ Binary search Tree.
- ✓ ~~Insertion, deletion, traversal~~
- ✓ Concepts of heap.
- ✓ Heap sort
- ✓ Trie
- ✓ Graph
- ✓ Graph travels (~~BFS~~, ~~DFS~~)
- ✓ Tree and its application.
- ✓ Complete, Full, perfect tree
- ✓ Bst $O(\log n)$.
- ✓ heap $O(\log n)$
- ✓ types of graph.
- ✓ isolated vertex
- ✓ weighted graph
- ✓ heap sort (~~$O(\log n)$~~) $O(n)$
- ✓ Trie vs hashtable.
- ✓ graph indexing
- ✓ edge indexing.
- ✓ Height vs depth of node.
- ✓ degree of node.
- ✓ Shortest path algorithm
- ✓ Binomial heap.

- ✓ BFS and DFS. Complexity
- ✓ Self balanced tree.
- ✓ top down and bottom up heapification.
- ✓ how set ~~BFS~~ bst is subset of X.
- ✓ adv. of priority queue using heap.
- ✓ adjacency list and adjacency matrix.
- ✓ Spanning tree.
- ✓ ~~Degree of node~~
- ✓ minimum spanning trees
- ✓ degree of graph
- ✓ Algorithm to find shortest path in graph.
- ✓ degree of all structures.
- ✓ Array tree
- ✓ Heapsort worst complexity?
- ✓ compressed trie.
- ✓ evaluation path attempt
- ✓ Dijkstra's algorithm
- ✓ Blatstone algo.

DS week -3 practicals.

- ✓ Find closet value to a given number in a tree?
- ✓ Validate wheather a given tree is BST or not?
- ✓ Find shortest distance b/w nodes in graph.
- ✓ Find word using prefix. t.m.c.
- ✓ Find Inf largest in vertex graph?
- ✓ Sorting - inorder - BST
- ✓ Search largest in BST
- ✓ Smallest parent in BST
- ✓ Find node with highest no. of edges
- ✓ Get all words match with given prefix
- ✓ Traverse DFS, BFS
- ✓ Smallest ancestor.
- ✓ Find shortest distance b/w vertices.

Practicals o + SQL

- Get list of employees with same salary.
- Create table with foreign key.
- Create transaction.
- delete row and rename column and delete column.
- change data type of column.
- getting largest name